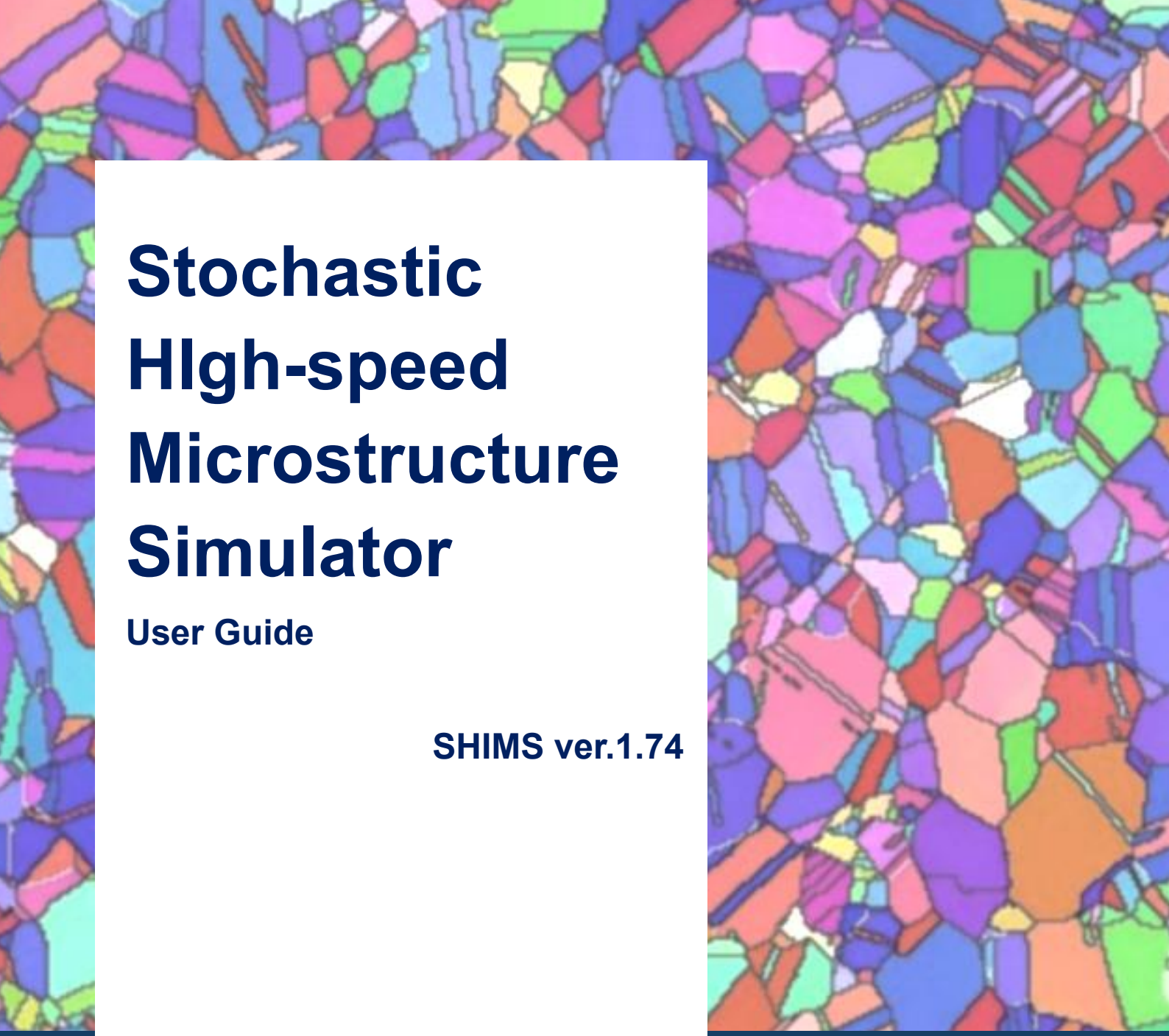

Stochastic HIgh-speed Microstructure Simulator

User Guide

SHIMS ver.1.74

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SHIMS User Guide

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1. Introduction

A. Aim of SHIMS

Prediction and control of microstructure evolution is significant for optimizing manufacturing processes for metallic products, from traditional methods to advanced technologies such as additive manufacturing (AM). However, representative computational simulation methods – *e.g.*, phase-field method – are challenging to apply in practical applications due to substantial computational cost.

SHIMS, a Stochastic HIgh-speed Microstructure Simulation, is an in-house microstructure simulation code based on the full-scale Monte Carlo Potts (MC Potts) models. SHIMS can provide accurate quantitative predictions in practical applications based on fundamental material properties, with high computational efficiency.

This document includes a brief overview of the SHIMS program, an explanation of the employed simulation models, an explanation of the functions of the SHIMS program, and examples.

전통적인 제조 기술부터 적층 제조와 같은 첨단 기술에 이르기까지, 금속 제품의 제조 공정을 최적화하기 위해서는 공정 중 미세구조 진화를 예측하고 제어하는 것이 매우 중요하다. 그러나 phase-field 모델로 대표되는 종래의 미세조직 시뮬레이션 기법들은 상당히 큰 계산 비용과 계산 시간을 요구하기 때문에 실용되기 어렵다.

SHIMS는 실제 크기 몬테카를로 포츠 모델 시뮬레이션을 위해 연구용으로 개발된 프로그램 코드이다. 재료의 기초적인 성질에 기반해 실제 공정들에 대해서도 정성·정량적인 측면에서 정확한 시뮬레이션 결과를 제공할 수 있으며, 높은 전산 효율을 보인다.

본 문서는 SHIMS 프로그램에 대한 개관, 활용된 몬테 카를로 모델들에 대한 설명, 프로그램 기능 설명, 그리고 실제 시뮬레이션 예시들을 포함하고 있다.

B. Main algorithm and parallel computing

SHIMS, a Stochastic HIgh-speed Microstructure Simulation, is an in-house code that provides three simulation modules: MC Potts grain growth simulation (GG module), finite difference method (FDM)-combined MC Potts simulation (SOL module), and additive manufacturing (AM) simulation (AM module).

In the GG module, an algorithm of the basic MC Potts model is active:

1. Randomly select a simulation voxel ("target site"). Calculate the total system free energy (E_1).
2. Randomly select one of the neighboring simulation voxels ("interacting site").
3. Replace the information of the target site with that of the interacting site. (This is a "trial".)
4. Calculate the total system free energy after a trial (E_2).
5. Calculate the acceptance probability of a trial (P) according to the system free energy change ($\Delta E=E_2-E_1$).
6. Generate a random number ($0 \leq \zeta \leq 1$) and compare it with the acceptance probability.
7. Accept the trial when $\zeta \leq P$, or reject the trial when $\zeta > P$.
8. Repeat the above steps.

In the SOL and AM modules, the MC Potts solidification model is active:

1. Randomly select a simulation voxel ("target site").
2. Randomly select one of the neighboring simulation voxels ("interacting site").
3. Change the information of the target site according to the type of interaction:
   3-1. (target) solid vs. (interacting) solid: the same as the grain growth model.
   3-2. (target) liquid vs. (interacting) liquid: insert new grain orientation into the target site (nucleation trial) and check whether a solid simulation voxel exists among the neighbors of the target site (to distinguish nucleation mechanism).
   3-3. (target) liquid vs. (interacting) solid: insert grain orientation information of the interacting site into the target site (epitaxial growth trial).
4. Calculate the acceptance probability of a trial (P).
5. Generate a random number ($0 \leq \zeta \leq 1$) and compare it with the acceptance probability.
6. Accept the trial when $\zeta \leq P$, or reject the trial when $\zeta > P$.
7. Repeat the above steps.

In addition, FDM is included in the algorithm for SOL and AM modules:

1. Determine Δt_{MCS} (the length of 1 MC step). and Δt_{FDM}^{max} (from the numerical limit).
2. If $\Delta t_{MCS} < \Delta t_{FDM}^{max}$, let Δt_{FDM}^{max} to be Δt_{MCS} and do one FDM step for each MCS (MC step).
   Otherwise, do N FDM steps for each MCS where N is the minimum integer that makes $\Delta t_{MCS} \geq N\Delta t_{FDM}^{max}$.

Schematic view of each simulation model and an overview of the simulation code are shown in Figs. 1 and 2.

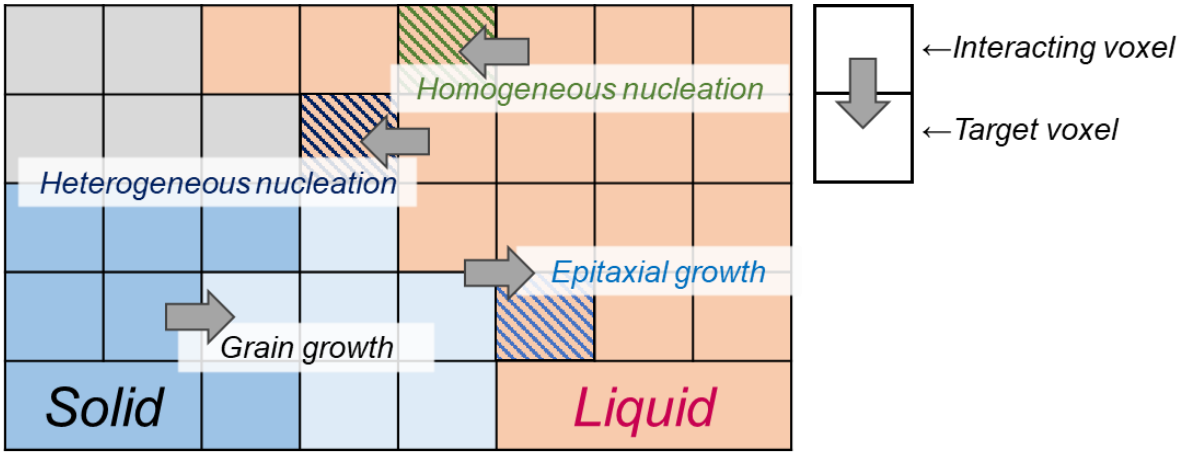


Figure 1. Schematic view of algorithms of MC Potts grain growth and solidification models.

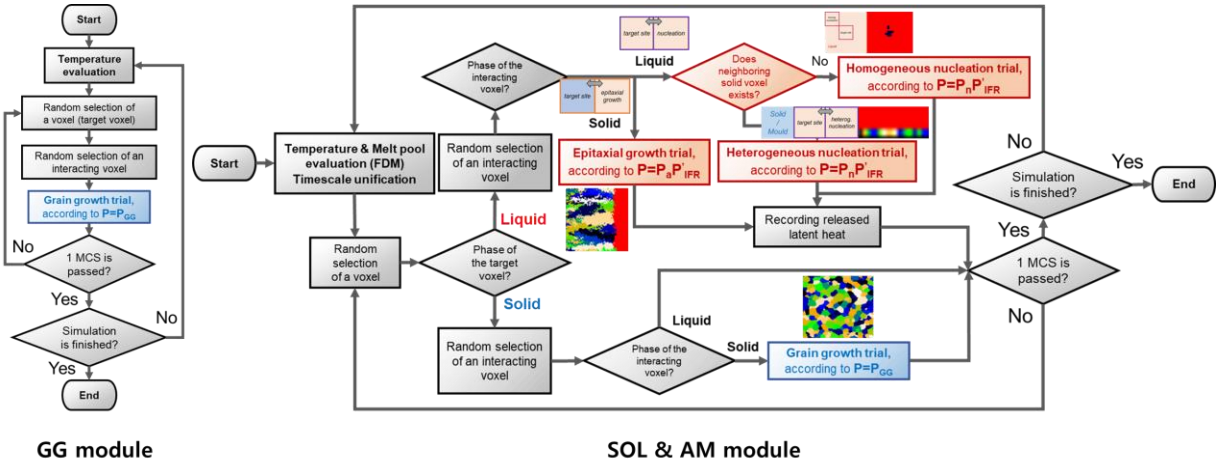


Figure 2. Schematic view of each simulation module.

SHIMS supports the message passing interface (MPI) for parallel computing. A schematic view of MPI processing is shown in Fig. 3.

The whole simulation domain is divided into several domains, and MC trials are attempted only for the assigned domain voxels at each processor. “Ghost voxels”, which provide the neighbor information required for MC trials for simulation voxels located near the domain boundaries (*i.e.*, boundary information), are required. In this case, the order of trial may cause a certain bias and error, since boundary information is not always shared due to computational efficiency.

To avoid such an error, each divided domain is once again divided into two sublattices (according to the direction of domain boundaries), one sublattice is randomly selected, trials corresponding to 1/2 MCS are attempted, boundary information is shared only in one side, trials corresponding to last 1/2 MCS are attempted, and finally the boundary information in is shared in other side.

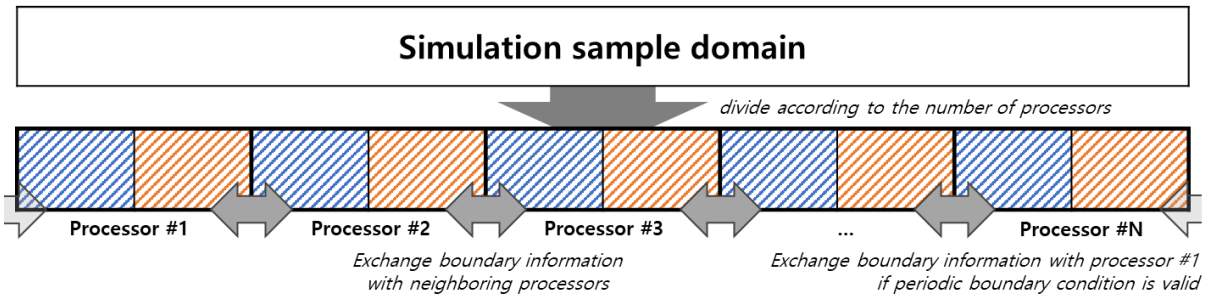


Figure 3. Schematic view of MPI in SHIMS. Different colors indicate different sublattices.

C. Monte Carlo Potts model for grain growth

The MC Potts model embedded in SHIMS employs a grain growth probability function which considers grain boundary anisotropy dependence and temperature dependence of grain growth [1]. The acceptance probability function of a grain growth trial is described as follows:

$$P = P_{mob}P_{therm} = \begin{cases} \exp\left[-\frac{Q}{k_B}\left(\frac{1}{T}-\frac{1}{T_{max}}\right)\right]\exp\left[-\frac{\Delta G}{k_B T}\right] & if\ \Delta G > 0 \\ \exp\left[-\frac{Q}{k_B}\left(\frac{1}{T}-\frac{1}{T_{max}}\right)\right] & if\ \Delta G \leq 0 \end{cases}$$

$$\Delta G = \Delta\left(\sum_i A_i\gamma_i\right) = dx^2\Delta\left(\sum_i \gamma_i\right)$$

where Q is the activation energy of grain boundary movement, k_B is the Boltzmann constant, T is the temperature, T_m is the melting temperature, γ_i is the grain boundary energy of interface i as a function of misorientation angle.

The time unit of the MC Potts simulation is Monte Carlo Step (MCS), defined as the ratio of the number of attempted trials to the total number of simulation voxels in the system. A realistic time unit is assigned to the MC Potts simulation by comparing the parabolic law of grain growth ($L^2=Kt$, where L is the average grain size, K is the growth rate constant) obtained from the simulation with that from the experiment [2, 3]:

$$\Delta t_{GG} = \Delta x^2 \frac{K_0^{MC}}{K_0^e} \cdot \Delta t_{MC}$$

where Δx is the size of a simulation voxel, K_0 is the pre-exponential factor for the growth rate constant $K = K_0 \exp(-Q_{GG}/k_B T)$ where Q_{GG} is the activation energy of grain growth, k_B is the Boltzmann constant, and T is the temperature. Δt_{MC} is the simulation time in MCS unit. In author's experience, simulation results are consistent independent of the size of a simulation voxel, as demonstrated in Fig. 4a.

It should be noted here that the physical meaning of Q is different from that of Q_{gg}. Q is the activation energy of GB migration (*i.e.*, purely for kinetics), while Q_{gg} includes both kinetic factors and thermodynamic factors for grain growth. Namely, they should be different.

Separating kinetic and thermodynamic factors is difficult in the real world. Atomistic simulations may be appropriate to obtain the exact value of Q, but it is not a simple work. On the other hand, a reasonable value can be estimated by making Q_{gg} in the simulation to be the same as that in the experiment, as demonstrated in Fig. 4b [3].

If the exact anisotropy of GB mobility should be modeled and applied to the simulation, then atomistic simulations would have to be *a priori* done. Consideration of exact GB anisotropy will be available in the future, while it cannot be in the present version code.

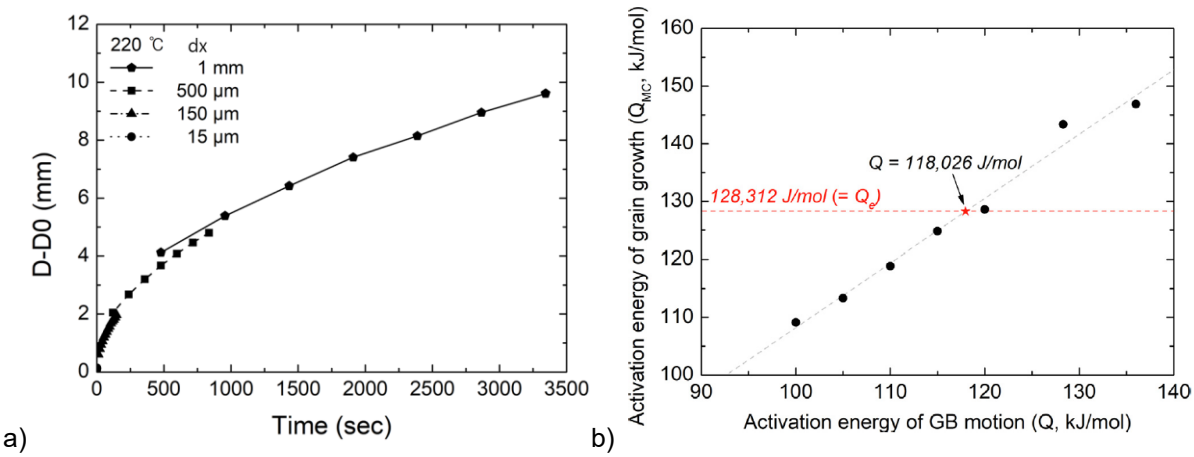


Figure 4. a) Average grain size in simulations with different sizes of a simulation voxel.
b) Activation energy of grain growth obtained from simulation results as a function of the activation energy of GB migration [3].

D. Monte Carlo Potts model for solidification

As already shown in Fig. 1, SHIMS considers epitaxial growth of existing grains and formation of new grains by homogeneous nucleation and heterogeneous nucleation. The acceptance probability function for interfacial reactions is [1, 4]:

$$P_{IFR} = \begin{cases} M\left(\frac{p_c^{max}}{p_c^{max}+p_s^{max}}\right)e^{-\frac{\Delta G}{k_B T}} & if\ \Delta G > 0 \\ M\left(\frac{p_c^{max}+p_s^{i}}{p_c^{max}+p_s^{max}}\right) & if\ \Delta G \leq 0 \end{cases}$$

M is the normalized mobility function of the interface, k_B is the Boltzmann constant, T is the temperature, ΔG is the free energy change due to an attempt in the simulation, p_c^i is the configurational contribution to the total free energy change (which depends on interface fluctuations), and p_s^i is the scalar contribution (which does not depend on interface fluctuation and is irreversibly applied) [1, 4].

Interface mobility is directly related to atomic diffusion at the interface. Atomic diffusion in liquid is fast, and its activation energy (*i.e.*, temperature dependence) is not as significant as that in solids. Accordingly, M is not expected to be significant to describe solidification reaction, so M was assumed to be 1. Then:

$$P'_{IFR} = \begin{cases} \left(\frac{p_c^{max}}{p_c^{max}+p_s^{max}}\right)e^{-\frac{\Delta G}{k_B T}} & if\ \Delta G > 0 \\ \left(\frac{p_c^{max}+p_s^{i}}{p_c^{max}+p_s^{max}}\right) & if\ \Delta G \leq 0 \end{cases}$$

The free energy change due to an attempt is expressed using the following equation:

$$\Delta G = -V\Delta G_{df} + \Delta\left(\sum A_i\gamma_i\right) = -dx^3\Delta G_{df} + dx^2\Delta\left(\sum \gamma_i\right)$$

where V is the volume, ΔG_{df} is the chemical driving force for phase transformation, A_i is the area of solid-liquid interface i, γ_i is the interfacial energy of interface i, and dx is the size of a simulation voxel.

For pure materials, the driving force ΔG_{df} during solidification mainly comes from thermal supercooling: $\Delta G_{df} = \Delta H_{fus}\,\Delta T/T_m$ where T_m is the melting temperature, ΔH_{fus} is the heat of fusion, and ΔT is the supercooling.

In contrast, for alloys, constitutional supercooling also plays a major role as a driving force during solidification. However, accounting for constitutional supercooling requires solving the diffusion problem, which requires enormous computational cost even in two-dimensional spaces.

Instead of additional consideration of compositional changes, the author introduced a simplified model to estimate the chemical driving force for alloy solidification. Assuming that the solute partitioning is not significant and long-range diffusion is limited (*e.g.*, due to a rapid cooling rate, as in AM processes), the initial liquid composition can be maintained during the solidification, and the initial solid composition can be assumed to be similar to that of the liquid. Then, ΔG_{df} can be modeled as a simple function of temperature (*e.g.*, $\Delta G_{df} = G_{Liquid} - G_{Solid} = a + b\,T + c\,T^2 + \cdots$). Here, the driving force is 0 at T_L and varies linearly within the solidification range, ensuring continuouity with the value of $\Delta G_{df}(= G_{Liquid} - G_{Solid})$ at T_S. A schematic illustration is shown in Fig. 5 [5].

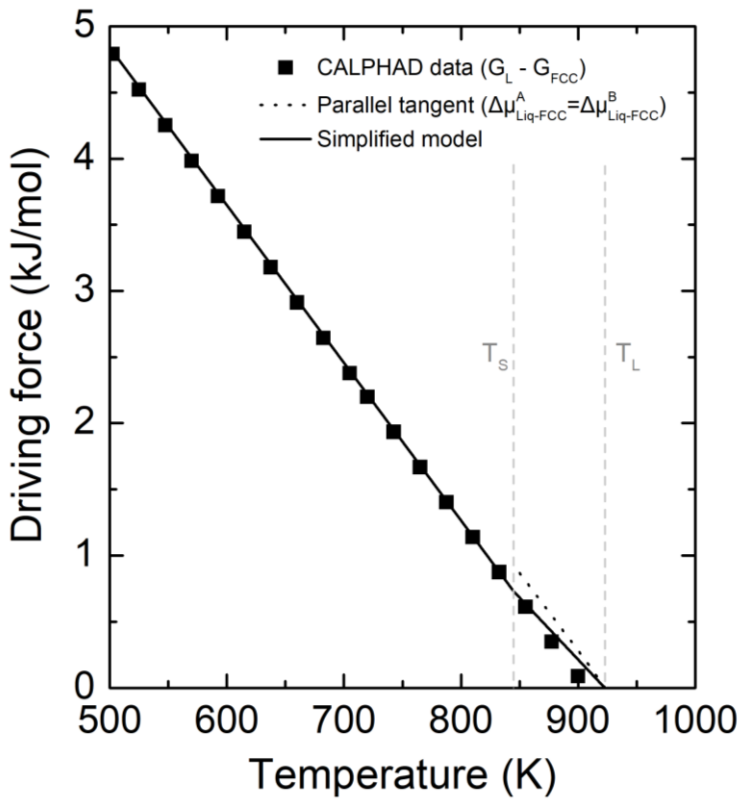


Figure 5. Estimated the chemical driving force for alloy solidification [5].

As the size of a simulation voxel increases to the μm scale, the volumetric driving force term ($V\Delta G_{df}$) becomes always dominant, making the acceptance probability independent of interfacial conditions. In such a case, the simulation cannot reflect the difference between nucleation and growth trials, since both will always be accepted.

To avoid such an artifact, an additional term for nucleation frequency (P_n) was newly introduced in the calculation of the acceptance probability, in addition to the existing term for growth probability: $P_{nucl} = P_n P'_{IFR}$. P_n depends on the number of nuclei that exceed the critical size within a simulation voxel. When the expected number of critical nuclei within a simulation voxel is less than 1, the simulation first tests whether a critical nucleus forms within the volume of a simulation voxel (N) and then whether the critical nucleus grows to the voxel size (P'_{IFR}). Otherwise, when multiple critical nuclei already exist in a simulation voxel, the simulation only tests whether one of them grows to the voxel size (P'_{IFR}). Namely :

$$P_n = \begin{cases} N & where\ \ N < 1 \\ 1 & where\ \ N \geq 1 \end{cases}$$

According to the Boltzmann distribution, the number of nuclei that exceed the critical size within a simulation voxel is $N = N_c \exp[-\Delta G^*/k_B T]$ where N_c is the number of nucleation sites and ΔG^* the critical nucleation energy barrier (*i.e.*, the free energy change associated with the formation of a nucleus with critical size). N_c was assumed to be $dx^3 N_{av}/V_m$ for homogeneous nucleation (where dx is the size of a simulation voxel, N_{av} is the Avogadro's number and V_m is the molar volume) and $dx^2 N_{av}/A_m$ (where A_m is the molar area) for the heterogeneous nucleation. The latter is assumed to be comparable to $(dx^3 N_{av}/V_m)^{2/3}$. According to the classical nucleation theory [6], ΔG^* is expressed as:

$$\Delta G^* = f(\theta) \frac{(\sum \gamma_i)^3}{\left(\Delta G_{df}\right)^2}$$

where $f(\theta)$ is geometric factor that depends on the shape of the nucleus. In the simulation, spherical homogeneous nuclei and lens-shaped heterogeneous nuclei at interface planes are assumed. In this case, $f(\theta)$ is expressed to be:

$$f(\theta) = \frac{16}{3}\pi \cdot \frac{(2+\cos\theta)(1-\cos\theta)^2}{4}$$

where θ is the wetting angle of a nucleus. θ is 180° for a homogeneous nucleus. During the simulation, the value of $f(\theta)$ is naturally determined by calculating the wetting angle (θ) of a heterogeneous nucleus from the Young's relation [7], as schematically presented in Fig. 6a [5].

Finally, the acceptance probability function for nucleation trial is:

$$P_{nucl} = \begin{cases} P_n \left(\frac{p_c^{max}}{p_c^{max} + p_s^{max}}\right) e^{-\frac{\Delta G}{k_B T}} & if\ \ \Delta G > 0 \\ P_n \left(\frac{p_c^{max} + p_s^{i}}{p_c^{max} + p_s^{max}}\right) & if\ \ \Delta G \leq 0 \end{cases}$$

For the growth trial, the formation of a protrusion does not create a new grain boundary plane. In this case, the wetting angle is calculated to be 0 according to Young's relation. Accordingly, the same acceptance probability as in nucleation trials (Eq. 10) could be naturally applied to growth trials, while P_n is assumed to be 1 for growth trials.

On the other hand, dendritic growth is typically directional, and the key factor determining dendrite growth direction is anisotropy in the solid-liquid interfacial energy [8]. Instead of sophisticated modeling to deal with crystallographic orientation as well as the degree of freedom of grain boundaries, a brief modeling is employed to consider directional growth.

The second nearest neighboring voxels in the Moore neighborhood, beyond the interacting voxel (in the same direction of interaction), are considered to calculate acceptance probability, by introducing an additional term (P_{aniso}). For epitaxial growth trials, it was assumed to be in the preferred growth direction if a solid voxel with the same grain orientation as the interacting solid voxel is located behind the interacting solid voxel. A schematic view of this algorithm is shown in Fig. 6b [5].

$$P_{aniso} = \begin{cases} 1 & for\ preferred\ growth\ direction \\ n & otherwise \end{cases}$$

Here, n is a user-defined number. Finally, the acceptance probability function for the growth trial is determined by the following equation:

$$P_{grow} = P_{aniso}P'_{IFR} = \begin{cases} \left(\dfrac{p_c^{max}}{p_c^{max} + p_s^{max}}\right) e^{-\frac{\Delta G}{k_B T}} & if\ \ \Delta G > 0 \\ \left(\dfrac{p_c^{max} + p_s^{i}}{p_c^{max} + p_s^{max}}\right) & if\ \ \Delta G \leq 0 \end{cases}$$

Theoretically, the temperature at the growing dendrite tip can increase during growth due to the release of latent heat, making a temperature gradient at the interface. Accordingly, the reasonable value of T may differ from a macroscopic perspective. To address this issue without losing computational efficiency, the simulation assigns a value of T randomly selected between T_∞ and T_{sol} when calculating the acceptance probability for each trial. Technically, T_∞ is the temperature of the liquid voxel attempting to solidify, and T_{sol} is the temperature of the interacting solid voxel.

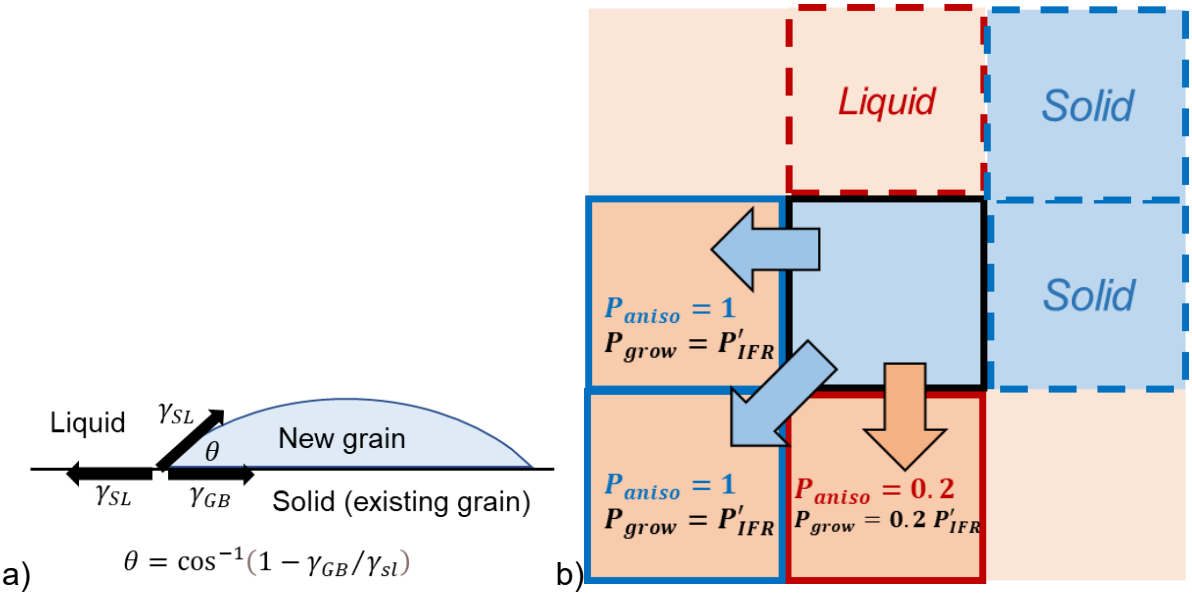


Figure 6. a) Calculation of the wetting angle for a heterogeneous nucleus [5].
b) Schematic view of the algorithm for growth direction anisotropy [5].

For solidification, the consideration of latent heat is important since thermal gradient plays a critical role in determining microstructure evolution [9]. Accordingly, it is necessary to solve the heat transfer equation in the simulation. In SHIMS, the finite difference method (FDM) is employed to account for heat transfer problems.

In FDM, the maximum length of timestep for an iteration is determined as $dx^2/2\alpha$

(where α is the thermal diffusion coefficient). On the other hand, one simulation step of the MC Potts model itself does not have a realistic time unit. To couple FDM with the MC Potts model using the given algorithm, a realistic time assignment for the MC Potts simulation is required.

Theoretically, dendrite tip velocity is expected to be proportional to the driving force (*e.g.*, thermal supercooling for pure materials [10, 11]):

$$v = \frac{\Delta x_{real}}{\Delta t_{real}} \cong K_{solidi}^{expt.} \Delta T$$

where K_i^{solidi} is the rate constant for solidification. Given that P_{grow} for the growth trial is proportional to ΔT (where $p_s^i = |V \Delta G_v| \propto \Delta T$), the growth velocity in the simulation is also expected to be linearly proportional to ΔT:

$$v = \frac{\Delta x_{MC}}{\Delta t_{MC}} \cong K_{solidi}^{MC} \Delta T$$

Comparing above equations, then a realistic timescale for the solidification model is:

$$\Delta t_{real}^{solidi} = \left(\frac{\Delta x_{real}}{\Delta x_{MC}}\right)\left(\frac{K_{MC}^{solidi}}{K_{real}^{solidi}}\right) \Delta t_{MC} = dx\, \beta\, \Delta t_{MC}$$

It has to be noted that the dendrite tip velocity is not linearly proportional to ΔT in the real world; rather, it is proportional to ΔT^3 [12-14]. This discrepancy may arise because the temperature does not remain perfectly isothermal during solidification due to the release of latent heat. When the latent heat release is involved in the simulation, such a behavior can be well reproduced [5]. Accordingly, a trial-and-error approach is required by adjusting the β parameter to reproduce the realistic solidification velocity in the simulation.

E. Time scale unification in complex simulation

When various MC models are simultaneously included in the simulation – *e.g.*, grain growth model and solidification model, which value should be chosen as the length of one MCS (Δt_{real}) among Δt_{real}^{solidi} for the solidification model and Δt_{real}^{gg} for the grain growth model?

The length of the timestep in the MC simulation can be adjusted by modifying the acceptance probability, according to the physics underlying the MC algorithm [15]. Using this scheme, a safe value of Δt_{real} without losing physical meaning can be determined as the shorter value between Δt_{real}^{solidi} and Δt_{real}^{gg}. In this case, each acceptance probability should be modified as follows:

$$P_{scale}^{i} = \frac{\Delta t_{real}}{\Delta t_{real}^{i}} P^{i}$$

where P^{i} is the acceptance probability for each reaction. With this scheme, the simulation can simultaneously consider various reaction kinetics without losing the physical meaning of each reaction kinetics [15]. An example is demonstrated in Fig. 7.

pure Ni, grain growth, 1200 K, 1 hour

Temperature (K)	1200			
Maximum acceptance probability, P_{max}	1.19×10^{-4}			
Actual length of 1 MCS, Δt (sec)	7.38×10^{-3}			
Acceleration factor, α	**1**	**10**	**100**	**1000**
The number of MCSs to simulate 1 hour of annealing	4.88×10^{5}	4.88×10^{4}	4.88×10^{3}	4.88×10^{2}
Average grain size after 1 hour of annealing (μm)	63±0.7	64±0.7	64±0.6	64±0.7

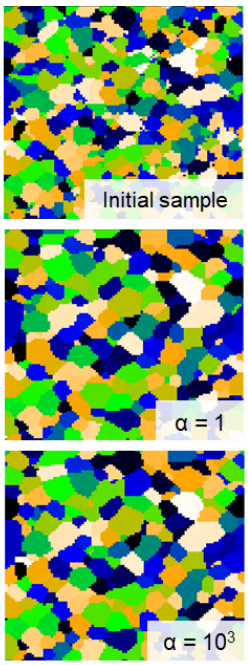


Figure 7. Grain growth simulation results, with different adjustments in timestep [15].

F. Finite difference method for heat transfer

The algorithm of the FDM-combined MC Potts simulation is described below:

1. Determine Δt_{FDM}^{max} from the numerical limit: $\Delta t_{FDM}^{max} = S\frac{dx^2}{2\alpha}$ where α is the thermal diffusivity ($= k/\rho c_p$), S is the safe factor for the numerical stability (~1/10 in SHIMS), and dx is the distance between grid points.
2. When $\Delta t_{MCS} > \Delta t_{FDM}^{max}$, do N FDM steps for each MCS, where N is the minimum integer that makes $\Delta t_{MCS} \geq N\Delta t_{FDM}^{max}$. Otherwise, define Δt_{FDM} to be Δt_{MCS} and do one FDM step per one MCS.

In 3D, FDM can be expressed as:

$$T_{t+\Delta t} = T + \frac{1}{\rho c_p}\frac{\delta Q}{\partial t}\Delta t + \frac{k\Delta t}{\rho c_p dx^2}\left[T_{x+1} + T_{x-1} + T_{y+1} + T_{y-1} + T_{z+1} + T_{z-1} - 6T\right]$$

where ρ is the density, c_p is the heat capacity, k is the thermal conductivity, and $\delta Q/\partial t$ is corresponding to the released latent heat. For the boundary condition, adiabatic, Robin (heat convection), and Dirichlet (fixed temperature) conditions are available in SHIMS.

The temperature recovery method [16] is utilized to account for latent heat release. In this case, the numerical solution from FDM changes to:

$$T_{t+\Delta t} = T + \delta T_{lat} + \frac{k\Delta t}{\rho c_p dx^2}\left[T_{x+1} + T_{x-1} + T_{y+1} + T_{y-1} + T_{z+1} + T_{z-1} - 6T\right]$$

The increment of temperature per a FDM step due to the latent heat release (δT_{lat}) is calculated to be $\Delta H_{fus}/(V_m c_p)$ where the unit for latent heat ΔH_{fus} is J/mol, molar volume V_m is mol/m^3, and heat capacity c_p is J/mol K. The effect of latent heat release should not be applied for a simulation voxel at once, not only for the numerical stability but also from a logical perspective [5]. Accordingly, the effect of latent heat release is assumed to gradually appear during the simulation, as follows:

1. Determine Δt from heat diffusion distance ($\sqrt{\alpha_h \Delta t}$, where α_h is the thermal diffusivity and Δt is the time).
2. When $\Delta t_{FDM} > \Delta t$, let $\delta T_{lat} = \frac{\Delta H_{fus}}{(V_m c_p)}$. Otherwise, let $\delta T_{lat} = \frac{\Delta H_{fus}}{(V_m c_p)}\frac{\Delta t_{FDM}}{\Delta t}$.

G. Additive manufacturing simulation

For the prediction of melt pool morphology and the temperature profile during the AM process, heating due to LASER irradiation should be modelled. The heat input from LASER irradiation is generally modeled using a Gaussian flux distribution:

$$F(x,y) = \frac{\alpha P}{2\pi\sigma_x\sigma_y}\exp\left[-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2}+\frac{y^2}{\sigma_y^2}\right)\right]$$

where σ_i is the standard deviation of the Gaussian in i direction, α is the LASER absorptivity, and P is the LASER power.

Simulating melt pool formation by applying a Gaussian flux requires a certain computational cost, which may even become greater than that for microstructure simulation, the primary objective of the SHIMS code. Even if a Gaussian flux is fully considered, the physical meaning of the predicted melt pool behavior still remains limited, as complicated modeling of the melt pool, *e.g.*, fluid dynamics inside the melt pool, is not considered.

In SHIMS, a simplified treatment is available as well as the Gaussian flux approach. In the simplified approach, the melt pool shape is defined by the user based on experimental observations, and the temperature at the melt pool boundary (*i.e.*, the interface between solid/powder and liquid) is set to the melting point of the material (or the liquidus temperature in the case of alloys). Around the melt pool, a temperature gradient is naturally formed through solving FDM for heat transfer. In both approaches, the latent heat absorption during melting was assumed to be compensated by laser irradiation and was therefore neglected.

For the Gaussian flux approach, the effect of LASER irradiation (acting as an external heat source) is considered during solving FDM, and then (before proceeding to an MCS) the simulation code checks whether any voxel has reached a temperature above the liquidus (melting point); such voxels are then converted into liquid.

For the simplified approach, the temperature of voxels within the defined melt pool region is set to the user-defined temperature (where a certain thermal gradient can also be set within the melt pool region). After solving FDM, an additional check is done to identify voxels above the liquidus temperature and to convert them into liquid accordingly.

2. Modules

In the main screen of SHIMS, four modules are available: Sample Maker, Simulation, Database, and Auxiliary Tools. Details of each module are described later in this chapter.

```
=================================================================
!  Stochastic HIgh-speed Microstructure Simulator               !
!                      based on the Monte Carlo Potts Model      !
!                                          SHIMS  ver.  1.71     !
!                                                                !
!      Coded by  S.-H. Oh                                        !
!                CMSE, POSTECH                                   !
!                shoh97@postech.ac.kr                            !
=================================================================


Select module:
  1 / m = Sample Maker
  2 / s = Monte Carlo Potts simulation
  3 / d = Database Checker
  4 / t = Auxiliary Tools: Graphics & Measurement
  0 / q = Quit
  >>
```

A. Sample Maker

In the Sample Maker module, the user can make initial microstructure samples with a cuboidal shape, multiply, crop, and rotate samples, and merge different samples. Note that all samples should be in a cuboidal shape in SHIMS.

To make a new sample, the options below are required to be input:

Input option	Description
Sample size (x, y, z)	Total size of the microstructure sample
Sample type	Solid: a solid sample full of grain(s) Liquid: a liquid sample Mould-liquid: a liquid sample with small grains at a boundary plane Powder: a sample filled with powder particles
Size of initial grains (Nx, Ny, Nz)	Size of initial grains with a cuboidal shape
File name	Name of created sample file * Name of file should not exceed 100 chars.

The “Multiply” option allows the sample to be replicated and expanded. The options below are required to be input:

Input option	Description
Input file name	Target file to replicate
Multiply factor (X, Y, Z)	Number of replicates in x/y/z direction(s) *Input should be an integer.
Apply new orientation	Replace the grain orientation in the newly created part of the sample or not
File name	Name of created sample file * Name of file should not exceed 100 chars.

The “Crop” option is to crop the sample into a cuboidal shape.

The “Merge” option is to merge two different samples. If the size of the attaching plane does not match, then SHIMS automatically replicates each sample (according to LCM).

The “Rotate” option is to rotate the sample with respect to the specific axis (x/y/z).

B. Simulation

SHIMS is driven by the input file (“input.in”). If there is no “input.in” file in the directory, it automatically creates an “example.in” file. The input file consists of several sections: SIMCARD, FDMCARD, FROZCARD, TEMPCARD, SOLCARD, and AMCARD. # is for annotation, and the below table shows required options for each simulation type (GG module: MC Potts grain growth simulation without FDM, SOL module: FDM-combined MC Potts simulation, AM module: additive manufacturing simulation). Each section is explained in detail later in this section.

Section	Grain growth (GG module)	FDM-MC (SOL module)	AM (AM module)
SIMCARD	O	O	O
FDMCARD		O	O
FROZCARD	△	△	△
TEMPCARD	O		
SOLCARD		O	O
AMCARD			O

For each section, options and corresponding descriptions are listed. Note that “mandatory” options should be included in the input file, while “default” values will automatically be used in the simulation even if the corresponding option is not included in the input file, and “requirement” means that the option will only be active when a specific option is active. For “mandatory” columns, “O” means the option is mandatory for every module, “△” means it is mandatory but can be ignored depending on other option(s), and “G/S/A” means it is mandatory for each module (GG, SOL, AM, respectively).

SIMCARD includes options essential for performing MC simulation.

Option	Description	Mandatory	Available keywords	Default	Requirement
MDB	Name of the database file for materials information *If MDB = NONE, then the user can manually input material information.	O			
SYS	Name of the system in DB *DB format is described later.	△			MDB != NONE
STR	Name of initial microstructure file	O			
DX	Length of one simulation voxel	O	value (m)		
MODE	Type of simulation: “GG” for GG module, “CONV” or “SOL” for SOL module, and “AM” for AM module	O	GG / CONV / SOL / AM		
REAL	Active or inactive realistic time assignment *It should be “REAL = Y“ for SOL and AM modules.	O	Y / N		
FDM	Include or exclude FDM	O	Y / N		REAL = Y
PBC	A three-digit number for the periodic boundary condition Each digit represents whether PBC is turned on or off along each direction. *For AM, the “000” option is automatically set.	O	[XYZ] 1 = on 0 = off		
CRI	Criteria to stop the simulation “MCS” for MCS, “SEC” for realistic time, “LIQ” for remaining liquid fraction in the simulation sample.	O	MCS / SEC / LIQ		
NSTEP	The total number of MCS or the maximum time to finish the simulation	O	integer (MCS) / time (s)		
NDATA	Interval time to record data on the “result.dat” file: Number of total MCS or time in simulation	O	integer (MCS) / time (s)		
NPRNT	Interval time to generate microstructure files: Number of total MCS or time in simulation	O	integer (MCS) / time (s)		

TEMPCARD includes options for the isothermal condition in the GG module.

Option	Description	Mandatory	Available keywords	Default	Requirement
T(1) T(2) ...	Temperature of the whole sample as a function of time *Do not leave space in a single equation ex) T(2) = 50*t+50 → OK T(2) = 50*t + 50 → Error	G	equation line		FDM = N
t(1) t(2) ...	Time range over which the T(N) equation is valid	G	integer (MCS) / time (s) / 0 to input infinity		
tUNIT	Unit of t(N)	G	MCS / SEC		
ACCL	Acceleration factor *ACCL * P_{max} should be ≤ 1	G	number	1.0	MODE = GG

FROZCARD includes options for frozen temperature approximation. Note that the temperature gradient is available only in the x direction.

Option	Description	Mandatory	Available keywords	Default	Requirement
FROZ	Active or inactive frozen temperature approximation		Y / N		FDM = N
TINT	Initial temperature at x=0	△	temperature (K)		
TGRA	Temperature gradient *A positive number means the temperature increases as the x coordinate increases.	△	number (K/voxel)		

FDMCARD includes options for performing FDM.

Option	Description	Mandatory	Available keywords	Default	Requirement
T0	Initial temperature of the whole simulation sample *Type the name of the temperature field file instead of a number, to load temperature field information from an external file.	S,A	Temperature (K) / external file name		REAL = Y
BXL BXU BYL BYU BZL BZU	Boundary condition for solving the heat transfer eq. ex) “BYL” means “lower-y boundary (y=0)”	S,A	A (Adiabatic) / C (Robin) / F (Dirichlet)		
TXL TXU TYL TYU TZL TZU	Temperature of the outer medium, for solving the heat transfer eq.	△	Temperature (K)		B[?][?] = C or F
hXL hXU hYL hYU hZL hZU	Heat transfer coefficient for the outer medium	△	Value (W m^{-2} K^{-1})		B[?][?] = C

SOLCARD includes options for performing FDM-MC.

Option	Description	Mandatory	Available keywords	Default	Requirement
TCS	Arbitrary critical supercooling where solidification becomes possible ex) When TCS = 10, solidification will be considered when $T \leq T_m$-10.		Temperature (K)	0	FDM = Y
LAT	Include or exclude latent heat release		Y / N	Y	
NUCL	Include or exclude formation of new grains (by nucleation trials)		Y / N	Y	REAL = Y
PTCL	Fraction of particles (which act as heterogeneous nucleation sites)		fraction ≤ 1.0	0	FDM = Y MODE = AM
PANG	Minimum wetting angle of heterogeneous nuclei at the surface of particles		angle (degree) < 180.0 °	0.0	
WXL WXU WYL WYU WZL WZU	Wetting angle of heterogeneous nuclei at the boundary surfaces of the simulation sample		angle (degree) < 180.0 °	180.0	NUCL = Y FDM = Y PBC OFF W[?][?] != C
TIME	Determination scheme for the actual time corresponding to 1 MCS		1: timescale for the grain growth model only 2: set the smallest timescale among various models 3: take maximum acceleration factors at every OPT MCSs	2	FDM = Y
OPT	MCS intervals to check maximum acceleration factors.		integer (MCS)	1	FDM = Y TIME = 3
tMAX	Maximum time corresponding to 1 MCS		time (sec)	86400	
tSCA	The maximum ratio between the time for the next MCS and the time for the present MCS		number (> 1.0)	1000	

AMCARD includes options for melt pool information in AM module.

Option	Description	Mandatory	Available keywords	Default	Requirement
DIR	LASER scan direction axis	A	X / Y		MODE = AM
POS	Initial position for LASER irradiation ex) POS = 25 25 → origin is set to be (25,25) of upper z-surface	A	xpos ypos		
VEL	LASER scan velocity	A	value (m/s)		
MMOD	Melt pool modeling CIRCLE: circular shape melt pool is assumed. TEAR: teardrop shape melt pool is assumed. GAUSS: LASER irradiation is considered in solving FDM, based on the Gaussian distribution.	A	Circle / Tear / Gaussian		
TMP	Assumed temperature at the melt pool boundaries		temperature (K)	T_m	MODE = AM MMOD != G
RAD	Radius of assumed melt pool at upper z-surface	△	radius (voxel)		MODE = AM MMOD = C
DEP	Depth of assumed melt pool from upper z-surface	△	depth (voxel)		MODE = AM MMOD !=G
GMP	T gradient within the melt pool		T gradient (K/voxel)	0	MODE = AM MMOD = C
GEO	Geometry factor for the shape of a teardrop		factor	1.0	MODE = AM MMOD = T
WID	Maximum width of the melt pool	△	width (voxel)		
CTL	Total length (cap + tail length) of the melt pool	△	length (voxel)		
SX	LASER standard deviation in x direction	△	stdev (m)		MODE = AM MMOD = G
SY	LASER standard deviation in y direction	△	stdev (m)		
POW	LASER power	△	power (W)		
ABS	Relative LASER absorptivity		absorptivity	1.0	

The simulation result has been recorded in the “result.dat” file. Time, average grain size, and fraction of liquid are recorded, while fraction of liquid is excluded for the GG module.

```
$ MCS    GrainSize(Voxels)     time(s)    GrainSize(m)    f_liquid
   0     1.00              0.000E+00            0.000       0.995000
 100     1.06              5.000E-05            0.000       0.990353
 200     1.14              1.000E-04            0.000       0.983445
 300     1.22              1.500E-04            0.000       0.976105
```

Once a simulation is done, SHIMS automatically create graphical files of microstructure files that have been generated during the simulation (and also graphical files of temperature field files, in SOL and AM modules). The SHIMS graphics module is based on the gnuplot 4.6 version. Details of graphical treatment are described in Section 2D.

C. Database

In the Database module, the user can check the materials information taken from the database file, or can make an example file for the database file.

```
### Parameters for the < Al4Cu > system ###

  1. Grain Growth

     GBE (Max) = 0.65 J/m2  where theta_coh <= 15.0'
     Q_GB = 98220.0 J/mol, Tmax = 923 K

                 2     K0_e = 1.63E-05 m2/s

     t (sec) = dx  * ----------------------- * t_MC (MCS)
                       K0MC = 1.16E+04 unit2/MCS

  2. Solid-Liquid Transformation

                                /
                     LIQ       /
                              / IFE_lv = 0.86 J/m2
       IFE_sl = 0.15 J/m2 /
      --------------------      VAP(GAS)
                          \
                     SOL   \ IFE_sv = 1.03 J/m2
                            \

     dH_fus = 10770.0 J/mol = 1.102354E+09 J/m3  where Vm = 9.77E-06 m3/mol
     Tm = 923 K, Tsuper = 1.99 * Tm = 1837 K
                       Ks_MC
     t (sec) = dx  * ------- ( = 5.00E-02 s unit/m MCS) * t_MC (MCS)
                       Ks_e

  3. Heat Conduction

  Thermal conductivity k = 238.0 W/m K
  Heat capacity Cp = 37.112 J/K-mol
  ==========================================================================
```

The database file format is shown here; SHIMS does not read lines that begin with “#” or lines that are written after “EOF”. The user can also generate an example file (example.sdb) by using the Database module. In the database file, the sequence of each parameter should be the same as shown below.

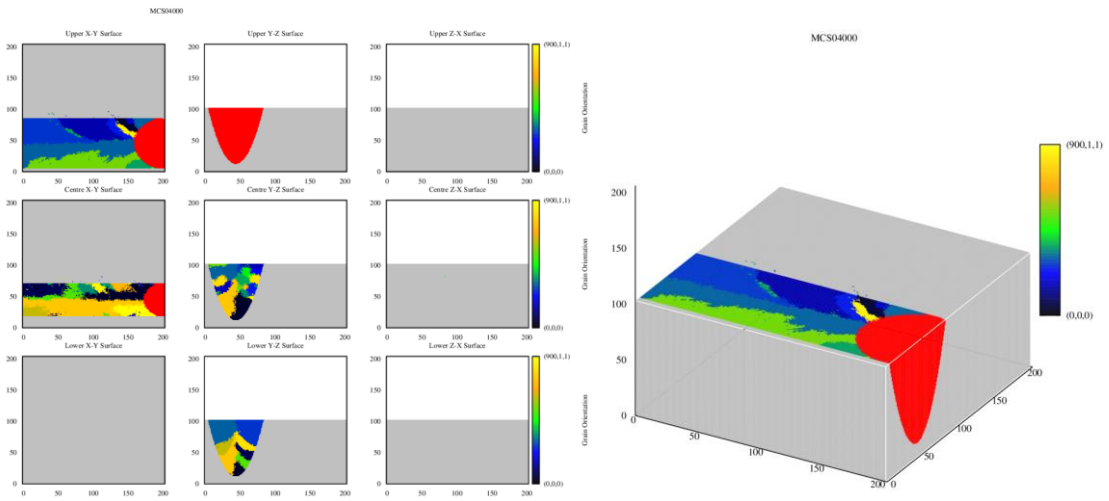
```
#NAME    Tliq    Tsol    GBEmax Aniso     Qmob       K0_e       K0_MC    beta_SL    dHfus    Vm_solid E_IFsl E_IFsv E_IFlv k_heat Cp(at_mp) IFE_aniso T0
fccNi     1728 1728    1.732          15 295140 0.05319    3925335      3.0E-3    17160    6.99E-6    0.356    2.104    1.750    88.5     38.14         0.2
Al4Cu       923    845      0.65          15    98220 1.63E-5    11627.6      5.0E-2    10770    9.77E-6    0.154    1.032    0.865     238     37.112        0.2            903
EOF
T(K) GBEmax(J/m2) Aniso(degree) Q(J/mol) D0(m2/s) K0_e(m2/s) K0_MC (site2/s) dHfus(J/mol) Vm (m3/mol-atom) E_IF(J/m2) k_heat (W/m K) Cp (J/K-mol)
```

Parameter	Note
Tliq Tsol T0	Melting temperature; Liquidus, solidus, and T0 temperature in K unit. *For pure materials, Tliq=Tsol and T0 is not necessary.
GBEmax	Maximum grain boundary energy, in J m^{-2} unit
Aniso	GBE anisotropy level for Read-Schockley equation.
Qmob	Activation energy for grain boundary mobility, in kJ mol^{-1} unit.
K0_e K0_MC	Pre-exponential term of grain growth rate constant, where L^2=Kt=K0 exp [-Q/RT] *K0_e for expt. data, and K0_MC for MC Potts simul. data
beta_SL	Time conversion constant for the solidification model
dHfus	Latent heat or heat of fusion, in J mol^{-1} unit
Vm_solid	Molar volume of solid, in m^3 mol^{-1} unit
E_IFsl	Solid-liquid interfacial energy, in J m^{-2} unit
E_IFsv	Solid-vapor interfacial energy, in J m^{-2} unit
E_IFlv	Liquid-vapor interfacial energy, in J m^{-2} unit
k_heat	heat conductivity, in W m^{-1} K^{-1} unit
Cp(at_mp)	Molar specific heat near melting temperature, in J K^{-1} mol^{-1} unit
IFE_aniso	Anisotropy factor for epitaxial growth direction

D. Auxiliary Tools & Visualization

In SHIMS, auxiliary functions are used to analyze microstructure and to visualize the microstructure and temperature field.

For visualization, two types of graphics files are generated: one file contains 2D cross-sections (yz, xz, and xy planes where x, y, z = 0, 1/2 max, and max, respectively), and another file contains a 3D overview of the sample. SHIMS code is developed based on gnuplot version 4.6, and the executable file named “gnuplot4” should be located at ~/bin/ directory. The user can also check the graphics at a specific position or visualize only a part of the sample.



Other functions provided by SHIMS are shown below:

Function	Description
Calculate avg. grain size	Calculate the average grain size based on the flood-fill algorithm *PBC is not considered.
Calculate interface position	Calculate the solid-liquid interface position *Only available in x direction
Calculated PDAS	Calculate avg. columnar grain thickness at each position *User can measure it in x, y, z direction.
Calculated avg. T	Calculate avg. temperature of the .tdat file
Analyze orientation fraction	Measure grain orientation across the sample *Information is recorded in the “Orient.dat” file.
Convert .dat / .tdat into .vtk	Convert the grain morphology file (.dat) or temperature field file (.tdat) into a .vtk file for VTK programs (*e.g.*, Paraview).

3. Examples

A. Hypothetical grain growth simulation

> cat input.in

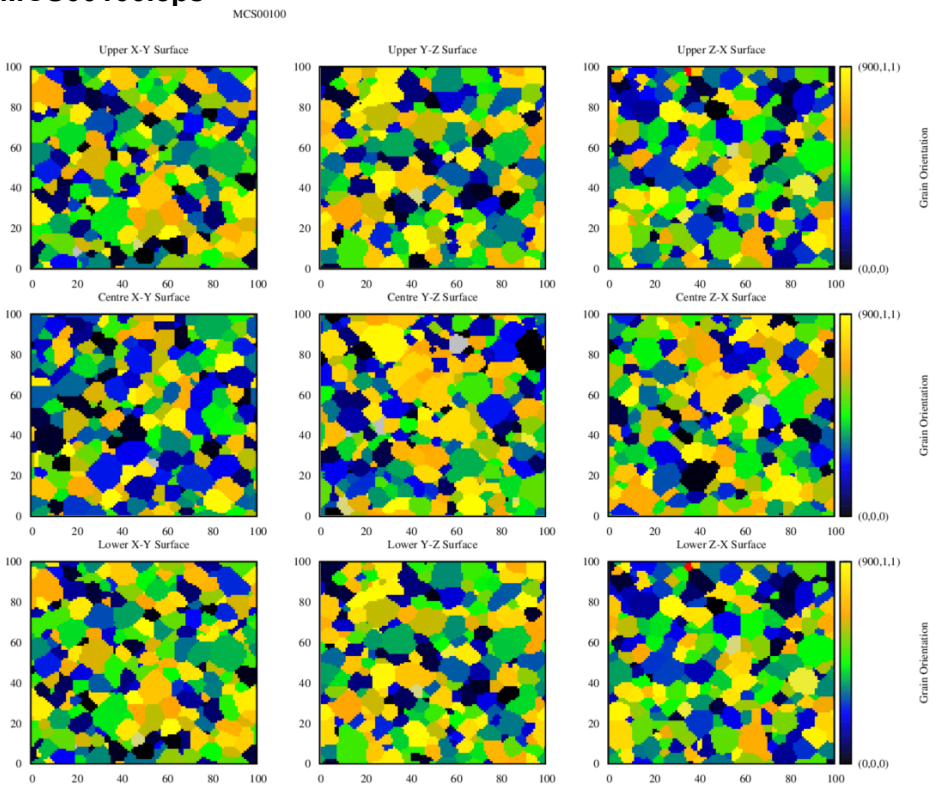
```
##########################################################################################
# SHIMS INCAR FORMAT
# YES = Y, NO = N
# USE "#" FOR ANNOTATIONS
# NO MORE THAN 100 CHAR IN ONE LINEs
##########################################################################################
# SIMCARD
MDB    = NONE
STR    = str.ini
DX     = 1E-6 m    ## Arbitrary value required for calculating free energy change
MODE = GG
REAL = N
FDM    = N
PBC    = 111      ## PBC for all direction
CRI    = MCS
NSTEP = 100 MCS
NDATA =    10 MCS
NPRNT =    10 MCS
##########################################################################################
# TEMPCARD (ONLY AVAILABLE WHEN FDM = N & FROZ = N)
T(1) = 900
t(1) = 0          # 0 for infinity time range
tUNIT = MCS
#EOF
```

```
Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 1

Edit mode? (1 = Make new / 2 = Multiply / 3 = Crop / 4 = Merge / 5 = Rotate // 0 = Quit )
  >> 1

Sample size (x direction, voxels)?: 100
Sample size (y direction, voxels)?: 100
Sample size (z direction, voxels)?: 100

Sample type? (0 = Solid, 1 = Liquid, 2 = Mould-liquid, 3 = Powder)
  >> 0

Size of initial grains? (1 grain will take Nx x Ny x Nz voxels)
  Nx = 1
  Ny = 1
  Nz = 1

Random Grain Created...
```

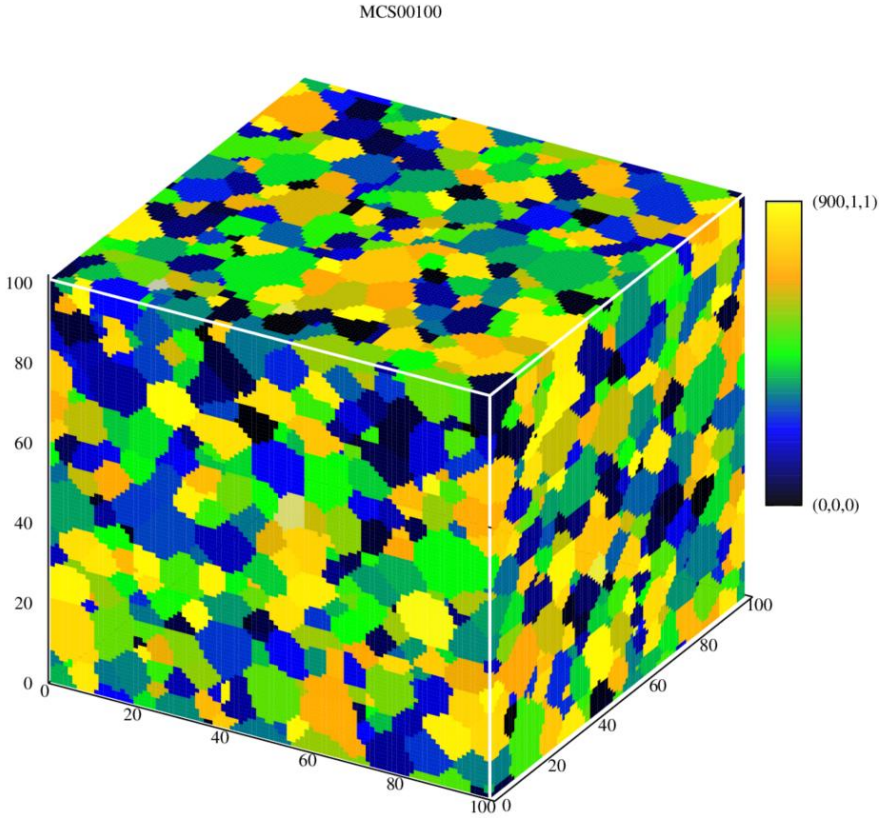
```
New sample file name?
 >> str.ini
      "str.ini" file was created...

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 2

Manual input for simulation parameters
Melting point of Material (liquidus temperature) (K) = 1000
@@ Arbitrary value
Max Grain Boundary Energy for Misorientation?(J/m2)
 >> 1.0
@@ Arbitrary value
GB Energy Anisotropy Level(Degree) = 15
@@ Arbitrary value, for Read-Schockley type anisotropy
Grain boundary mobility energy barrier? (J/mol)
 Q = 10000
@@ Arbitrary value

### NOTE: heat transfer model requires real time conversion...
                Automatically disabled...
   # Initial setting input is completed! #

Load input file "str.ini"...
 ( Site = 100 x 100 x 100 ) successfully loaded...!!!
      "MCS00000.dat" file was created...
####### MODULE : GG #########

 << Monte Carlo Simulation Started >>
      Sample size : 100 X 100 X 100
      Total MCS 100 steps, Print Grain Map every 10 steps, Print Result every 10 steps
      Single core mode

 # New T(t) function accepted: T(t) = 900 after t = 0 MCS
 10th MC Step was taken...
      "MCS00010.dat" file was created...
 20th MC Step was taken...
      "MCS00020.dat" file was created...
 30th MC Step was taken...
      "MCS00030.dat" file was created...
 40th MC Step was taken...
      "MCS00040.dat" file was created...
 50th MC Step was taken...
      "MCS00050.dat" file was created...
 60th MC Step was taken...
      "MCS00060.dat" file was created...
 70th MC Step was taken...
      "MCS00070.dat" file was created...
 80th MC Step was taken...
      "MCS00080.dat" file was created...
 90th MC Step was taken...
```

```
      "MCS00090.dat" file was created...
  100th MC Step was taken...
      "MCS00100.dat" file was created...

Execution time: 0 hour 2 min 39 sec
== Simulation is done ==

Load input file "MCS00000.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00010.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00020.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00030.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00040.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00050.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00060.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00070.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00080.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00090.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Load input file "MCS00100.dat"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 0
```

> display MCS00100.eps



> display 3d_MCS00100.eps



B. Realistic grain growth simulation

> cat input.in

```
##########################################################################################
# SHIMS INCAR FORMAT
# YES = Y, NO = N
# USE "#" FOR ANNOTATIONS
# NO MORE THAN 100 CHAR IN ONE LINEs
##########################################################################################
# SIMCARD
MDB    = material.sdb
SYS    = fccNi
STR    = str.ini
DX     = 10E-6 m    # “m” is just memo of the user; DX is always in “m” unit
MODE = GG
REAL = Y
FDM    = N
PBC    = 111
CRI    = MCS
NSTEP = 100 MCS
NDATA =    10 MCS
NPRNT =    10 MCS
##########################################################################################
# TEMPCARD (ONLY AVAILABLE WHEN FDM = N & FROZ = N)
T(1) = 1600    # unit K
t(1) = 0
tUNIT = MCS
ACCL = 1.0     # Acceleration factor if the user want to promote the timescale
#EOF
```

```
Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 1

Edit mode? (1 = Make new / 2 = Multiply / 3 = Crop / 4 = Merge / 5 = Rotate // 0 = Quit )
  >> 1

Sample size (x direction, voxels)?: 100
Sample size (y direction, voxels)?: 100
Sample size (z direction, voxels)?: 100

Sample type? (0 = Solid, 1 = Liquid, 2 = Mould-liquid, 3 = Powder)
  >> 0

Size of initial grains? (1 grain will take Nx x Ny x Nz voxels)
  Nx = 1
  Ny = 1
  Nz = 1

Random Grain Created...
New sample file name?
  >> str.ini
```

```
      "str.ini" file was created...

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 1
 # Initial setting input is completed! #

Load input file "str.ini"...
  ( Site = 100 x 100 x 100 ) successfully loaded...!!!
      "MCS00000.dat" file was created...

####### MODULE : GG #########

  << Monte Carlo Simulation Started >>
      Sample size : 100 X 100 X 100
      Total MCS 100 steps, Print Grain Map every 10 steps, Print Result every 10 steps
      1 MCS = 7.379836E-03 sec
      Single core mode

  # New T(t) function accepted: T(t) = 1600 after t = 0 MCS
  10th MC Step was taken...
      "MCS00010.dat" file was created...
  20th MC Step was taken...
      "MCS00020.dat" file was created...
  30th MC Step was taken...
      "MCS00030.dat" file was created...
  40th MC Step was taken...
      "MCS00040.dat" file was created...
  50th MC Step was taken...
      "MCS00050.dat" file was created...
  60th MC Step was taken...
      "MCS00060.dat" file was created...
  70th MC Step was taken...
      "MCS00070.dat" file was created...
  80th MC Step was taken...
      "MCS00080.dat" file was created...
  90th MC Step was taken...
      "MCS00090.dat" file was created...
  100th MC Step was taken...
      "MCS00100.dat" file was created...

Execution time: 0 hour 4 min 10 sec
== Simulation is done ==

Load input file "MCS00000.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00010.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00020.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!
```

```
Load input file "MCS00030.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00040.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00050.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00060.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00070.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00080.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00090.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00100.dat"...
  ( Site = 50 x 50 x 50 ) successfully loaded...!!!

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 0
```

> cat result.dat

```
$ MCS      GrainSize(Voxels)      time(s)      GrainSize(m)
    0        1.00                0.000E+00                0.010
   10        1.19                7.380E-02                0.012
   20        1.49                1.476E-01                0.015
   30        1.86                2.214E-01                0.019
   40        2.24                2.952E-01                0.022
   50        2.57                3.690E-01                0.026
   60        2.83                4.428E-01                0.028
   70        3.03                5.166E-01                0.030
   80        3.21                5.904E-01                0.032
   90        3.36                6.642E-01                0.034
  100        3.50                7.380E-01                0.035
```

C. Ingot casting simulation

> cat input.in

```
##########################################################################################
# SHIMS INCAR FORMAT
# YES = Y, NO = N
# USE "#" FOR ANNOTATIONS
# NO MORE THAN 100 CHAR IN ONE LINEs
##########################################################################################
# SIMCARD
MDB    = material.sdb
SYS    = fccCu
STR    = str.ini
DX     = 50E-6
MODE = CONV
REAL = Y
PBC    = 011
CRI    = LIQ
NSTEP = 1000
NDATA =     50
NPRNT =     50
##########################################################################################
# FDMCARD
T0     = 1358 K
FDM    = Y
BXL    = F    # Dirichlet condition @ lower x-surface
TXL    = 900 K
BXU    = A    # Adiabatic condition @ upper x-surface
BZL    = A    # Adiabatic condition @ lower z-surface
BZU    = C    # Robin condition @ upper z-surace
TZU    = 300 K # medium temperature
hZU    = 300 W/m2 K # heat transfer coefficient
##########################################################################################
# FROZCARD (ONLY AVAILABLE WHEN FDM = N)
FROZ = N
##########################################################################################
# SOLCARD
TCS = 0
LAT    = Y
NUCL = Y
WXL    = 0.0
WXU    = 180.0
WZL    = 180.0
WZU    = 180.0
TIME = 3    # optimization to maximum timescale
tMAX = 1E-2 sec # maximum length of 1 MCS
tSCA = 10 # maximum ratio; t_now/t_prev
#EOF
```

> mpirun -np 4 SHIMS

```
Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 1
```

```
Edit mode? (1 = Make new / 2 = Multiply / 3 = Crop / 4 = Merge / 5 = Rotate // 0 = Quit )
  >> 1

Sample size (x direction, voxels)?: 100
Sample size (y direction, voxels)?: 50
Sample size (z direction, voxels)?: 50

Sample type? (0 = Solid, 1 = Liquid, 2 = Mould-liquid, 3 = Powder)
  >> 1
Liquid box created...

New sample file name?
  >> str.ini
      "str.ini" file was created...

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 2

  # Initial setting input is completed! #

Load input file "str.ini"...
  ( Site = 100 x 50 x 50 ) successfully loaded...!!!
      "MCS00000.dat" file was created...

####### MODULE : SOL #########
      "MCS00000.tdat" T file created...
  ### NOTE: EVEN IN THE ACCELERATION MODE, FIRST 1 MCS IS IN MINIMUM MODE... ###

  << Monte Carlo Simulation Started >>
      Sample size : 100 x 50 x 50
      Total MCS 1000 steps, Print Grain Map every 50 steps, Print Result every 50 steps
      1 MCS = 1.250000E-07 sec, Latent heat for S-L reaction is considered
      t_gg    = 1.970186E-01 sec, t_solidi = 1.250000E-07 sec, t_FDM = 1.250000E-07 sec
      Execute 1 FDM loops for 1 MCS
      4 CPUs: MPI activated

  50th MC Step was taken...
      "MCS00050.dat" file was created...
      "MCS00050.tdat" T file created...
  100th MC Step was taken...
      "MCS00100.dat" file was created...
      "MCS00100.tdat" T file created...
  150th MC Step was taken...
      "MCS00150.dat" file was created...
      "MCS00150.tdat" T file created...
  200th MC Step was taken...
      "MCS00200.dat" file was created...
      "MCS00200.tdat" T file created...
  250th MC Step was taken...
      "MCS00250.dat" file was created...
      "MCS00250.tdat" T file created...
```

```
 300th MC Step was taken...
     "MCS00300.dat" file was created...
     "MCS00300.tdat" T file created...
 350th MC Step was taken...
     "MCS00350.dat" file was created...
     "MCS00350.tdat" T file created...
 400th MC Step was taken...
     "MCS00400.dat" file was created...
     "MCS00400.tdat" T file created...
 450th MC Step was taken...
     "MCS00450.dat" file was created...
     "MCS00450.tdat" T file created...
 500th MC Step was taken...
     "MCS00500.dat" file was created...
     "MCS00500.tdat" T file created...
 550th MC Step was taken...
     "MCS00550.dat" file was created...
     "MCS00550.tdat" T file created...
 600th MC Step was taken...
     "MCS00600.dat" file was created...
     "MCS00600.tdat" T file created...
 650th MC Step was taken...
     "MCS00650.dat" file was created...
     "MCS00650.tdat" T file created...
 700th MC Step was taken...
     "MCS00700.dat" file was created...
     "MCS00700.tdat" T file created...
 750th MC Step was taken...
     "MCS00750.dat" file was created...
     "MCS00750.tdat" T file created...
 800th MC Step was taken...
     "MCS00800.dat" file was created...
     "MCS00800.tdat" T file created...
 850th MC Step was taken...
     "MCS00850.dat" file was created...
     "MCS00850.tdat" T file created...
 900th MC Step was taken...
     "MCS00900.dat" file was created...
     "MCS00900.tdat" T file created...
 950th MC Step was taken...
     "MCS00950.dat" file was created...
     "MCS00950.tdat" T file created...
 1000th MC Step was taken...
     "MCS01000.dat" file was created...
     "MCS01000.tdat" T file created...

Execution time: 0 hour 2 min 49 sec
== Simulation is done ==

Load input file "MCS00000.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00050.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00100.dat"...
```

```
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00150.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00200.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00250.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00300.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00350.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00400.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00450.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00500.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00550.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00600.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00650.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00700.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00750.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00800.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00850.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00900.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS00950.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!

Load input file "MCS01000.dat"...
 ( Site = 100 x 50 x 50 ) successfully loaded...!!!
```

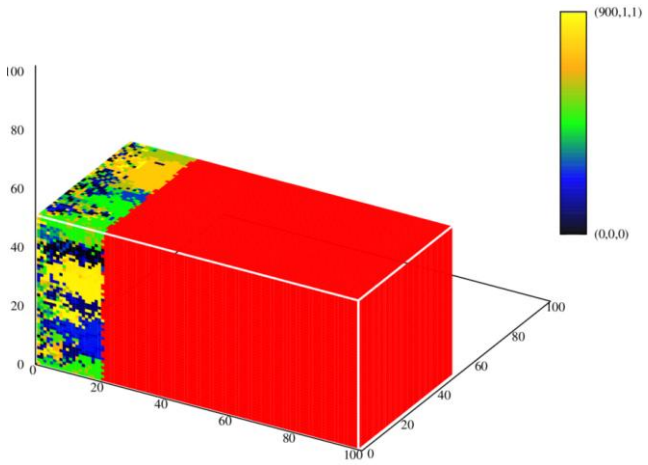
```
Draw T distribution? (GNUPLOT program is required, y=1 / n=0)
 >> 1
Load input file "MCS00000.tdat"...
Successfully loaded...!!!

Load input file "MCS00050.tdat"...
Successfully loaded...!!!

Load input file "MCS00100.tdat"...
Successfully loaded...!!!

Load input file "MCS00150.tdat"...
Successfully loaded...!!!

Load input file "MCS00200.tdat"...
Successfully loaded...!!!

Load input file "MCS00250.tdat"...
Successfully loaded...!!!

Load input file "MCS00300.tdat"...
Successfully loaded...!!!

Load input file "MCS00350.tdat"...
Successfully loaded...!!!

Load input file "MCS00400.tdat"...
Successfully loaded...!!!

Load input file "MCS00450.tdat"...
Successfully loaded...!!!

Load input file "MCS00500.tdat"...
Successfully loaded...!!!

Load input file "MCS00550.tdat"...
Successfully loaded...!!!

Load input file "MCS00600.tdat"...
Successfully loaded...!!!

Load input file "MCS00650.tdat"...
Successfully loaded...!!!

Load input file "MCS00700.tdat"...
Successfully loaded...!!!

Load input file "MCS00750.tdat"...
Successfully loaded...!!!

Load input file "MCS00800.tdat"...
Successfully loaded...!!!

Load input file "MCS00850.tdat"...
Successfully loaded...!!!
```

```
Load input file "MCS00900.tdat"...
Successfully loaded...!!!

Load input file "MCS00950.tdat"...
Successfully loaded...!!!

Load input file "MCS01000.tdat"...
Successfully loaded...!!!

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 0
```

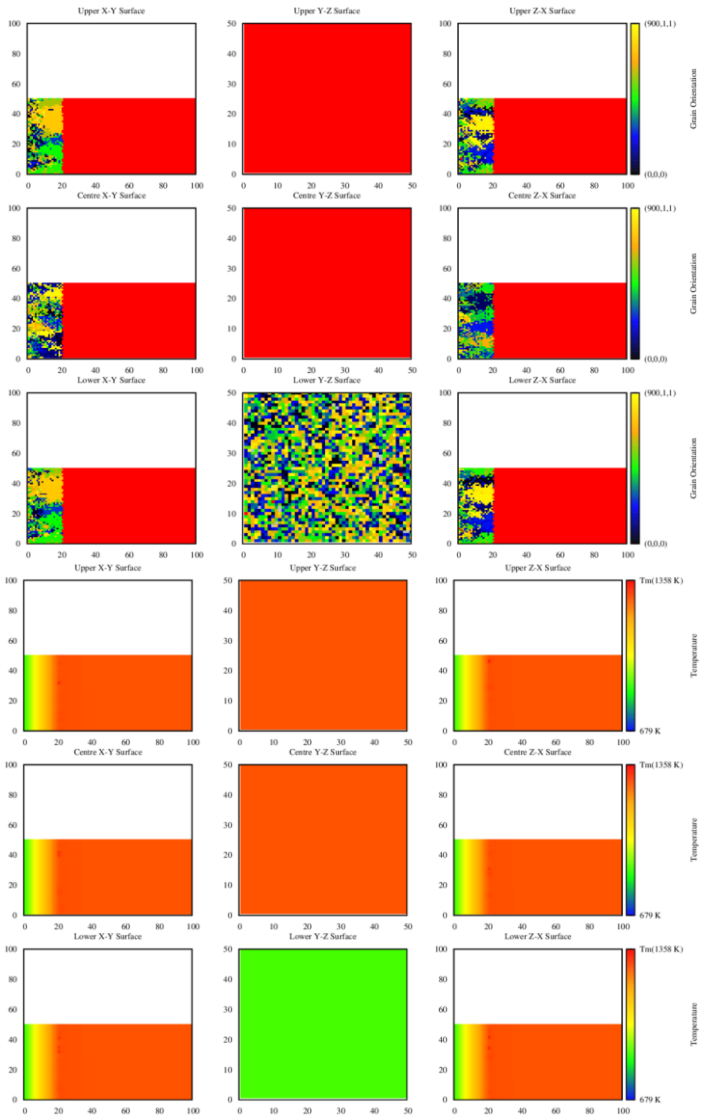
> cat result.dat

```
$ MCS    GrainSize(Voxels)    time(s)      GrainSize(m)    f_liquid
    0    0.00                 0.000E+00          0.000      1.000000
   50    1.10                 3.966E-05          0.000      0.980328
  100    1.12                 1.108E-04          0.000      0.973184
  150    1.16                 2.145E-04          0.000      0.961484
  200    1.19                 3.412E-04          0.000      0.950548
  250    1.21                 4.781E-04          0.000      0.940612
  300    1.23                 6.495E-04          0.000      0.930932
  350    1.24                 8.057E-04          0.000      0.922708
  400    1.26                 9.844E-04          0.000      0.914376
  450    1.27                 1.201E-03          0.000      0.905752
  500    1.29                 1.469E-03          0.000      0.895780
  550    1.31                 1.754E-03          0.000      0.886312
  600    1.33                 2.107E-03          0.000      0.876236
  650    1.35                 2.455E-03          0.000      0.866376
  700    1.37                 2.830E-03          0.000      0.856824
  750    1.39                 3.236E-03          0.000      0.847132
  800    1.41                 3.672E-03          0.000      0.837332
  850    1.43                 4.180E-03          0.000      0.826832
  900    1.44                 4.706E-03          0.000      0.816652
  950    1.46                 5.248E-03          0.000      0.806576
 1000    1.48                 5.911E-03          0.000      0.795176
```

> display 3d_MCS01000.eps



> display MCS01000.eps; display Tmap_MCS01000.eps



D. Additive manufacturing simulation

> cat input.in

```
##########################################################################################
# SIMCARD
MDB   = material.sdb
SYS   = Al4Cu
STR   = str.ini
DX    = 10E-6
MODE = AM
REAL = Y
PBC   = 000
CRI   = LIQ
NSTEP = 1000
NDATA =    50
NPRNT =    50
##########################################################################################
# FDMCARD
T0    = 300.0
FDM   = Y
BXL   = F
TXL   = 300.0
BXU   = F
TXU   = 300.0
BYL   = F
TYL   = 300.0
BYU   = F
TYU   = 300.0
BZL   = F
TZL   = 300.0
BZU   = C
TZU   = 300.0
hZU   = 25.0
##########################################################################################
# FROZCARD (ONLY AVAILABLE WHEN FDM = N)
FROZ = N
##########################################################################################
# SOLCARD (ONLY AVAILABLE WHEN REAL = Y)
TCS   = 0
LAT   = Y
NUCL = Y
PTCL = 0
PANG = 0.0
WXL   = 180.0
WXU   = 180.0
TIME = 3
tMAX = 1E-2
tSCA = 2
OPT   = 1
##########################################################################################
# AMCARD (ONLY AVAILABLE WHEN REAL = Y AND MODE = AM)
DIR   = X
POS   = 50 50
VEL = 600E-3 m/3
MMOD = C
RAD = 50
DEP   = 50
TMP   = 923
#EOF
```

```
> mpirun -np 4 SHIMS

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 1
Edit mode? (1 = Make new / 2 = Multiply / 3 = Crop / 4 = Merge / 5 = Rotate // 0 = Quit )
  >> 1
Sample size (x direction, voxels)?: 200
Sample size (y direction, voxels)?: 100
Sample size (z direction, voxels)?: 50

Sample type? (0 = Solid, 1 = Liquid, 2 = Mould-liquid, 3 = Powder)
  >> 3

Size of powder particles? (1 powder particle will take Nx x Ny x Nz voxels)
  Nx = 2
  Ny = 2
  Nz = 2

Random Powder Pool Created...

New sample file name?
  >> str.ini
       "str.ini" file was created...

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 2

  # Initial setting input is completed! #

Load input file "str.ini"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!
       "MCS00000.dat" file was created...

####### MODULE : AM #########
       "MCS00000.tdat" T file created...
  ### NOTE: EVEN IN THE ACCELERATION MODE, FIRST 1 MCS IS IN MINIMUM MODE... ###

  << Monte Carlo Simulation Started >>
       Sample size : 200 X 100 X 50
       Total MCS 1000 steps, Print Grain Map every 50 steps, Print Result every 50 steps
       1 MCS = 5.000000E-07 sec, Latent heat for S-L reaction is considered
       t_gg    = 7.133497E-02 sec, t_solidi = 5.000000E-07 sec, t_FDM = 7.980183E-08 sec
       Execute 6 FDM loops for 1 MCS
       Circular shaped melt pool
       # Simulation will automatically stop when there is no liquid... #
       4 CPUs: MPI activated
```

```
50th MC Step was taken...
    "MCS00050.dat" file was created...
    "MCS00050.tdat" T file created...
100th MC Step was taken...
    "MCS00100.dat" file was created...
    "MCS00100.tdat" T file created...
150th MC Step was taken...
    "MCS00150.dat" file was created...
    "MCS00150.tdat" T file created...
200th MC Step was taken...
    "MCS00200.dat" file was created...
    "MCS00200.tdat" T file created...
250th MC Step was taken...
    "MCS00250.dat" file was created...
    "MCS00250.tdat" T file created...
300th MC Step was taken...
    "MCS00300.dat" file was created...
    "MCS00300.tdat" T file created...
350th MC Step was taken...
    "MCS00350.dat" file was created...
    "MCS00350.tdat" T file created...
400th MC Step was taken...
    "MCS00400.dat" file was created...
    "MCS00400.tdat" T file created...
450th MC Step was taken...
    "MCS00450.dat" file was created...
    "MCS00450.tdat" T file created...
500th MC Step was taken...
    "MCS00500.dat" file was created...
    "MCS00500.tdat" T file created...
550th MC Step was taken...
    "MCS00550.dat" file was created...
    "MCS00550.tdat" T file created...
600th MC Step was taken...
    "MCS00600.dat" file was created...
    "MCS00600.tdat" T file created...
650th MC Step was taken...
    "MCS00650.dat" file was created...
    "MCS00650.tdat" T file created...
700th MC Step was taken...
    "MCS00700.dat" file was created...
    "MCS00700.tdat" T file created...
750th MC Step was taken...
    "MCS00750.dat" file was created...
    "MCS00750.tdat" T file created...
800th MC Step was taken...
    "MCS00800.dat" file was created...
    "MCS00800.tdat" T file created...
850th MC Step was taken...
    "MCS00850.dat" file was created...
    "MCS00850.tdat" T file created...
900th MC Step was taken...
    "MCS00900.dat" file was created...
    "MCS00900.tdat" T file created...
950th MC Step was taken...
```

```
      "MCS00950.dat" file was created...
      "MCS00950.tdat" T file created...
  1000th MC Step was taken...
      "MCS01000.dat" file was created...
      "MCS01000.tdat" T file created...

Execution time: 0 hour 14 min 24 sec
== Simulation is done ==

Load input file "MCS00000.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00050.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00100.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00150.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00200.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00250.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00300.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00350.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00400.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00450.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00500.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00550.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00600.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00650.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00700.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00750.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!
```

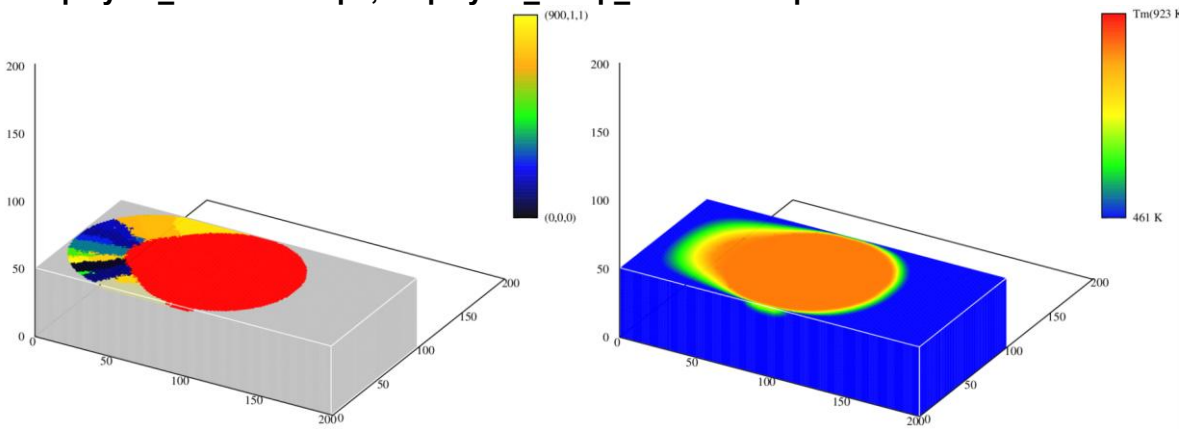
```
Load input file "MCS00800.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00850.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00900.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS00950.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Load input file "MCS01000.dat"...
  ( Site = 200 x 100 x 50 ) successfully loaded...!!!

Draw T distribution? (GNUPLOT program is required, y=1 / n=0)
  >> 1
Load input file "MCS00000.tdat"...
Successfully loaded...!!!

Load input file "MCS00050.tdat"...
Successfully loaded...!!!

Load input file "MCS00100.tdat"...
Successfully loaded...!!!

Load input file "MCS00150.tdat"...
Successfully loaded...!!!

Load input file "MCS00200.tdat"...
Successfully loaded...!!!

Load input file "MCS00250.tdat"...
Successfully loaded...!!!

Load input file "MCS00300.tdat"...
Successfully loaded...!!!

Load input file "MCS00350.tdat"...
Successfully loaded...!!!

Load input file "MCS00400.tdat"...
Successfully loaded...!!!

Load input file "MCS00450.tdat"...
Successfully loaded...!!!

Load input file "MCS00500.tdat"...
Successfully loaded...!!!

Load input file "MCS00550.tdat"...
Successfully loaded...!!!

Load input file "MCS00600.tdat"...
Successfully loaded...!!!
```

```
Load input file "MCS00650.tdat"...
Successfully loaded...!!!

Load input file "MCS00700.tdat"...
Successfully loaded...!!!

Load input file "MCS00750.tdat"...
Successfully loaded...!!!

Load input file "MCS00800.tdat"...
Successfully loaded...!!!

Load input file "MCS00850.tdat"...
Successfully loaded...!!!

Load input file "MCS00900.tdat"...
Successfully loaded...!!!

Load input file "MCS00950.tdat"...
Successfully loaded...!!!

Load input file "MCS01000.tdat"...
Successfully loaded...!!!

Select module:
   1 / m = Sample Maker
   2 / s = Monte Carlo Potts simulation
   3 / d = Database Checker
   4 / t = Auxiliary Tools: Graphics & Measurement
   0 / q = Quit
   >> 0
```

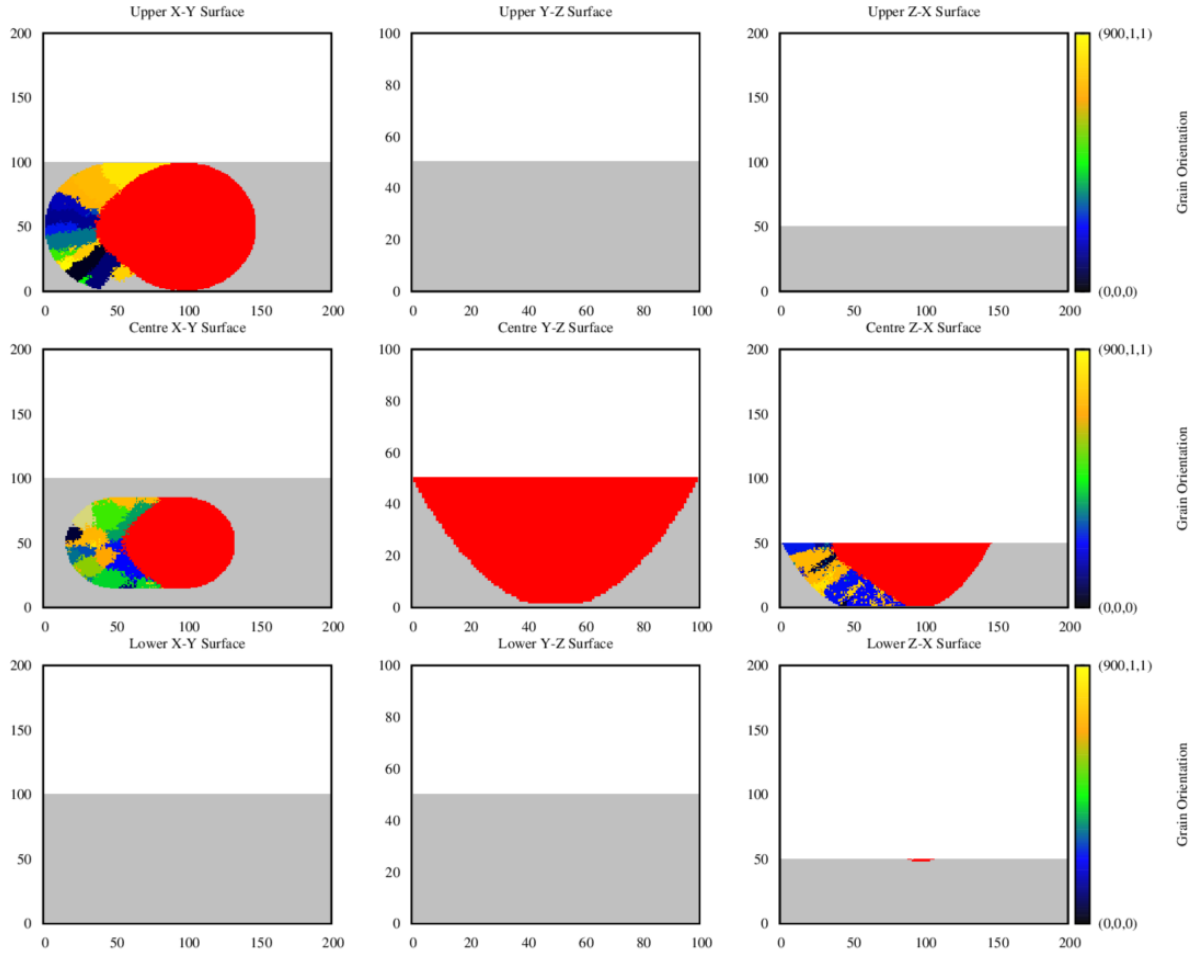
> cat result.dat

$ MCS	GrainSize(Voxels)	time(s)	GrainSize(m)	f_liquid
0	2.00	0.000E+00	2.0E-05	0.000000
50	1.99	3.778E-05	2.0E-05	0.206358
100	1.99	6.945E-05	2.0E-05	0.212664
150	1.99	9.963E-05	2.0E-05	0.217985
200	1.99	1.276E-04	2.0E-05	0.221332
250	2.00	1.580E-04	2.0E-05	0.217721
300	2.00	1.966E-04	2.0E-05	0.217416
350	2.01	2.385E-04	2.0E-05	0.213638
400	2.01	2.798E-04	2.0E-05	0.214213
450	2.02	3.225E-04	2.0E-05	0.211777
500	2.03	3.652E-04	2.0E-05	0.213092
550	2.03	4.069E-04	2.0E-05	0.211499
600	2.04	4.496E-04	2.0E-05	0.213204
650	2.04	4.921E-04	2.0E-05	0.211892
700	2.05	5.349E-04	2.0E-05	0.213739
750	2.05	5.758E-04	2.1E-05	0.216203
800	2.06	6.186E-04	2.1E-05	0.214631
850	2.07	6.603E-04	2.1E-05	0.216948
900	2.07	7.020E-04	2.1E-05	0.215542
950	2.08	7.446E-04	2.1E-05	0.217565
1000	2.09	7.867E-04	2.1E-05	0.216247

> display 3d_MCS01000.eps; display 3d_Tmap_MCS01000.eps



> display MCS01000.eps

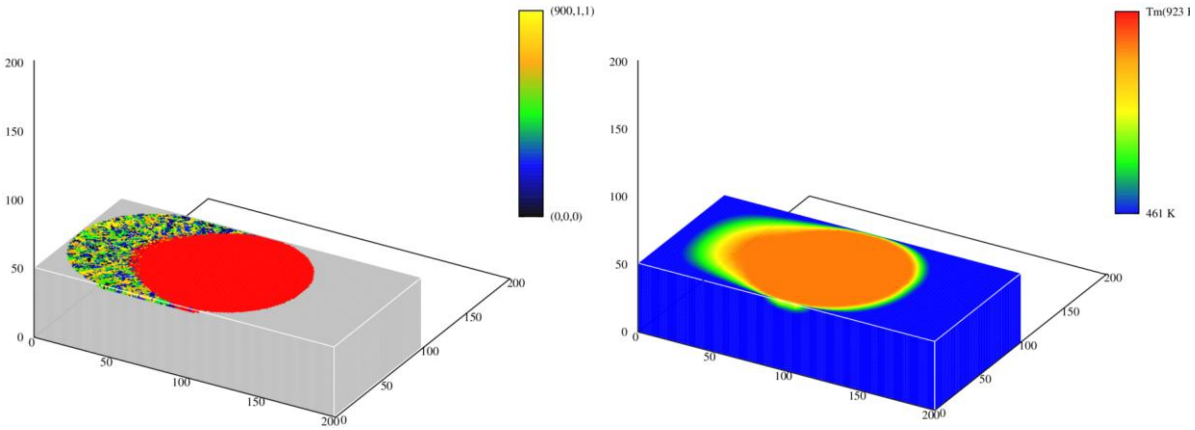


E. Additive manufacturing with particles

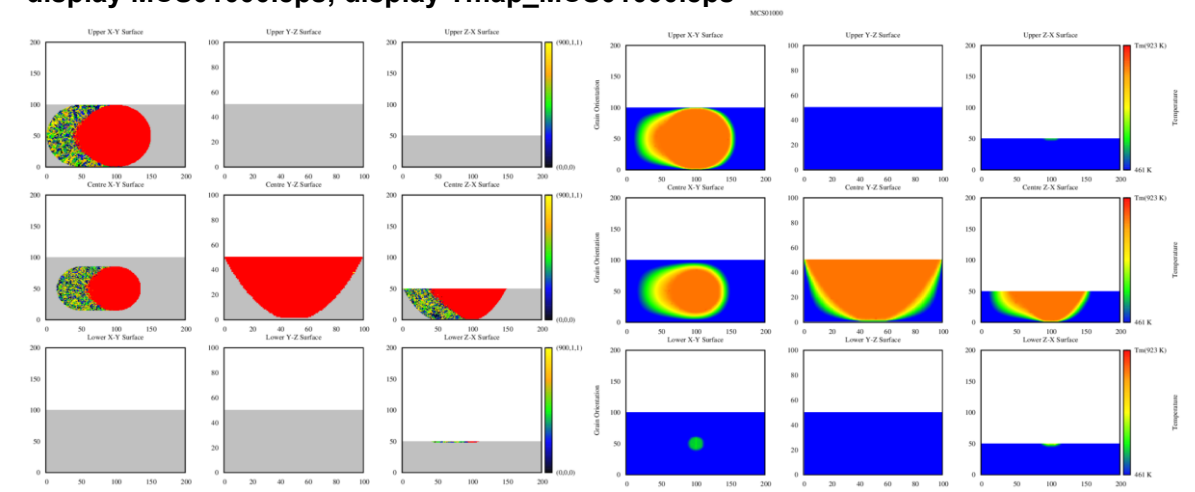
> cat input.in

```
##########################################################################################
# SHIMS INCAR FORMAT
# YES = Y, NO = N
# USE "#" FOR ANNOTATIONS
# NO MORE THAN 100 CHAR IN ONE LINEs
##########################################################################################

                                                    .
                                                    .
                                                    .

# SOLCARD (ONLY AVAILABLE WHEN REAL = Y)
PTCL = 0.1    # IMC particle fraction of 10 % within liquid region
PANG = 0.0    # Minimum wetting angle of heterogeneous nuclei, which determines nucleation energy barrier

                                                    .
                                                    .
                                                    .

#EOF
```

> display 3d_MCS01000.eps; display 3d_Tmap_MCS01000.eps



> display MCS01000.eps; display Tmap_MCS01000.eps



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